THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A road is 2.7 km. A runner has run 1850 m. The remaining distance is: 1

(a) 650 m **(b)** 850 m **(c)** 750 m **(d)** 1850 m

2. In the experiment above, the worst conductor is: 1

(a) S **(b)** P **(c)** R **(d)** Q

3. Hot water at 80 °C is poured into three cups: metal, ceramic and plastic. After 5 minutes the water is coolest in the: 1

(a) ceramic cup **(b)** all equal **(c)** plastic cup **(d)** metal cup

4. A student heats 100 g of ice and records temperatures. The graph has a flat portion at 0 °C and another at 100 °C. The flat portions represent: 1

(a) heating of solid and liquid **(b)** freezing
(c) melting and boiling **(d)** cooling

5. In the mixture above, after the first step the residue on the filter paper is: 1

(a) iron **(b)** salt **(c)** water **(d)** sand

6. A student observes a crystal growing in a saturated solution. Which statement is true? 1

(a) the crystal grows by outside deposition, not from within
(b) it reproduces
(c) it respires
(d) it is living

7. In that experiment, water rises in the jar by about one fifth because: 1

(a) nitrogen forms one-fifth **(b)** water is heavy
(c) oxygen forms about one-fifth of air **(d)** jar is wide

8. A student draws the Moon at the same time each night and sees the bright part grows. The Moon is: 1

(a) waxing **(b)** new **(c)** missing **(d)** waning

9. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) type of plant **(b)** amount of light **(c)** amount of soil **(d)** amount of water

10. Why do prey animals like deer have eyes on the sides of the head? 1

(a) wide field of view to spot predators **(b)** to dig
(c) to see only forward **(d)** to swim

11. A person eats only steamed vegetables and fruits and nothing else. The likely lack is: 1

(a) water **(b)** vitamins **(c)** protein and fat **(d)** fibre

12. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron and aluminium **(b)** iron only

(c) wood **(d)** all metals

13. Which of the following is true about a freely falling stone? 1

(a) it is in linear motion and its speed increases

(b) it is at rest

(c) it moves in a circle

(d) it is in periodic motion

14. If material C is an opaque metal and D is opaque wood, how would you distinguish them? 1

(a) C conducts heat and D does not **(b)** both float

(c) both transparent **(d)** both dissolve

15. If a 20 °C object touches a 60 °C object, which gets warmer? 1

(a) the 20 °C object **(b)** the 60 °C object **(c)** both cool **(d)** neither

16. If 50 mL of water evaporates from a vessel in 5 days, the average loss per day is: 1

(a) 250 mL **(b)** 5 mL **(c)** 10 mL **(d)** 50 mL

17. **Assertion (A):** Winnowing is used to separate heavier grains from lighter husk. 1

Reason (R): Wind blows away the lighter husk while the heavier grains fall down.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** A stone is a non-living thing. 1

Reason (R): A stone does not grow, breathe or respond to changes around it.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** Sunlight is a renewable resource. 1

Reason (R): It is available every day and does not get used up by our use.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The Sun is a star. 1
Reason (R): The Sun produces its own light and heat.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. How can a scientific model such as a globe help us? 2

22. Why does a dolphin look like a fish but is classified as a mammal? 2

23. Why is a diet rich only in proteins and fats not balanced? 2

24. A student says that heating a magnet strengthens it. Do you agree? 2

25. Convert 0.003 km into cm. 2

26. Why do we choose rubber for the handle of an electrician's tools? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Describe an experiment to measure the length of a curved line of 1 metre length marked on a floor by two students. 3

28. Explain the importance of non-conducting handles in electrical tools. 3

29. Describe why the lid of a tiffin box should be airtight to keep food warm. 3

30. Why do hill stations often receive more rainfall? 3

31. Describe why a mixture of water and kerosene separates into two layers. 3

OR

Why can't we say that a seed is dead because it does not respire visibly? 3

32. A school discards 200 sheets of paper daily. If both sides were used, how many sheets would be needed? 3

OR

Explain how the Sun's gravity keeps planets in orbit and what would happen if it vanished. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Sunita measured the length of her classroom using her footsteps and got 12 steps. Her friend Priya measured the same room and got 15 steps. They were confused, so their teacher in Bhopal asked them to use a measuring tape. The tape gave 9 m for both girls. 5
(i) Name the non-standard unit used by the girls.
(ii) Why did they get different values?
(iii) What is the SI unit of length?
(iv) Express 9 m in centimetres and explain why a measuring tape is better.

34. **Case study.** Mansi's mother in Nagpur has a steel kadhai with a wooden handle, a glass bowl, a cloth bag and a plastic bucket. She showed that the glass bowl lets light through clearly, tracing paper lets some light through, and a wooden board lets none. 5

(i) Name the material of the kadhai.

(ii) Which material from the story is transparent?

(iii) Which is opaque?

(iv) Why is the handle of a kadhai made of wood and the kadhai of steel?

35. Explain how a solar salt pan produces salt in stages and how impurities are removed. 5

36. Discuss the importance of responses to stimuli in the survival of plants and animals with five examples. 5

OR

Discuss with examples how our daily lifestyle can be made more 'earth-friendly'. 5

37. Explain, using the Earth-Moon-Sun positions, why the same Moon phase can be seen all over the world on the same day but at different times. 5

OR

Three students measured the time of 20 pendulum swings: 28 s, 30 s, 32 s. Find the average, the time for one swing, and say why 20 swings are timed. 5

38. Describe how the structure of a bird's body (including beak, feet, bones, feathers) is adapted for its way of life with examples. 5

OR

Plan a balanced day's menu for a 12-year-old, justify each meal's nutrients and mention what to avoid. 5

END OF QUESTION PAPER **Total: 80 Marks**

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